

Lesson 0004 — Why the model fails on the robot

~75–100 MIN

Deployment gap

ANCHORED TO: [EVALUATION/M5_LIVE/REPORT.MD](#) · [EVALUATION/M4C_GEOMETRY_FILTER/REPORT.MD](#)

A strong score on one image distribution does not guarantee reliable detections in a different camera domain. We will localize the gap using the project's measured evidence.

HOW TO USE THIS LESSON

Read the explanation, inspect the visual, make a prediction, and then compare it with the repository evidence. The goal is a model of this project that you can explain—not a list of framework slogans.

PREREQUISITES AND SCOPE

This lesson assumes the training/validation split and basic precision/recall vocabulary from Lesson 0003. It teaches diagnosis, not a predetermined verdict: a live failure can originate in the data distribution, preprocessing, artifact, decoder, ROS wiring, or geometry filter.

Why the model fails on the robot

VALIDATION ≠ DEPLOYMENT · LOCALIZE THE FAILURE

READING ORDER

1 → 2 → 3 → ... → 12

1 TWO WORLDS



The model learns one image distribution; the JetRover supplies another distribution.

2 VALIDATION WINDOW



M2 validation contains only 9 images and 21 instances, so its estimate is noisy.

3 M4B SAVED FRAMES



Each class ID appeared in all 30 frames; coverage, not accuracy. Extra detections also appeared.

4 M5 LIVE ROOM

The downward room view shows smaller faces, different lighting, floor texture, and new context.

5 CANDIDATES VS KEEPS



A candidate can disappear before geometry; a rejected candidate is a different failure.

6 THRESHOLD TEST



Lowering confidence to 0.25 unlocked only 2 green keeps out of 439 messages.

7 DEPTH CANNOT RECOVER



The geometry filter cannot rescue red or blue when YOLO never proposes their boxes.

8 CONTROLLED EVIDENCE



Empty and cubes bags separate false positives from missed positives.

9 RATE IS SEPARATE



M5 publish rate was about 15.2–15.9 Hz; rate and detection correctness are different metrics.

10 STICKER HYPOTHESIS

Removing the sticker changed the zero-keep result by no meaningful amount.

11 HONEST VERDICT



M5 is PARTIAL: the node runs, the filter rejects floor texture, but room cubes are missed.

12 NEXT EXPERIMENT



Fine-tune on JetRover-room positives, then repeat the same acceptance gates.

PROJECT ANCHOR

anchors: evaluation/m5_live/report.md · models/README.md · docs/m3d-revived-plan.md

EXPLANATORY MAP · NOT A MEASUREMENT

Orientation map of the diagnosis path. The detailed sections add definitions, competing hypotheses, and evidence boundaries. Click the PNG to open it at full resolution; the source SVG remains available for inspection.

Retrieval warm-up

Answer from memory before revealing the cues. Difficulty here is useful: it strengthens retrieval and shows what deserves review.

1. What different jobs do the training, validation, and test sets perform?

► **Reveal recall cues**

2. What does “best checkpoint” mean—and what does it not prove?

► **Reveal recall cues**

The question this lesson answers

Why can a model score strongly on its validation images, work on one saved robot-camera set, and then miss the cubes in the live JetRover room? Because “the model” is only one part of a distribution-dependent pipeline. We need to compare the image domains and localize the stage at which evidence disappears.

BY THE END, YOU SHOULD BE ABLE TO

- explain why a validation metric applies to a sample rather than every future image;
- separate measured facts, interpretations, hypotheses, and untested possibilities;
- locate the first pipeline boundary where expected evidence disappears;
- design a controlled diagnostic that changes one suspected cause;
- state the M5 PARTIAL result without blaming an unproven single root cause.

PROJECT TRUTH

The current engineering status is **M5 PARTIAL**. The node builds, TensorRT loads, and the recorded empty-scene run has zero kept detections. The live cubes-in-frame gate is not met, and the geometry predicate's physical sign is wrong. These observations define the current boundary; they do not prove one exclusive root cause.

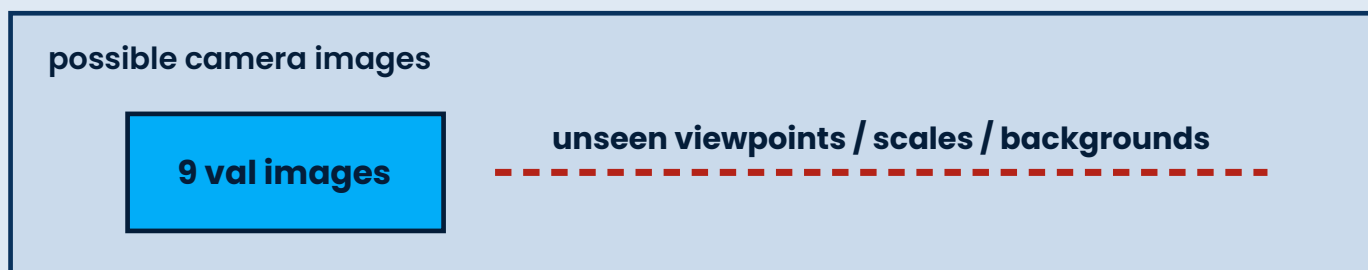
THREE SIMILAR IDEAS THAT ARE NOT SYNONYMS

A **data distribution** is the recurring mix of appearances and conditions that produce examples. A **domain shift** occurs when deployment conditions differ materially from training or validation. **Overfitting** occurs when a model fits available training evidence without learning patterns that generalize. A deployment gap may involve either or both, so evidence is needed before naming the cause.

The detector receives pixel arrays, not physical cubes. “The same cube” can therefore become a very different input when it is smaller, darker, blurred, viewed from above, or surrounded by unfamiliar objects. Domain shift is a change in the statistical input conditions; it does not require the physical object category to change.

1. Validation is a window, not the whole world

The M2 validation result was measured on 9 images and 21 instances from the Roboflow export. Its overall mAP@0.5 was 0.954. This is a valid statement about that validation sample. It is not the same as “95.4% accurate on every JetRover frame,” especially when the sample is small and visually different from deployment.



A metric is only as representative as the data behind it. The live room test asks a different question.

Sampling matters twice: nine images create statistical uncertainty, and visual mismatch creates representativeness uncertainty. Repeating evaluation on the same nine images reduces neither. The remedy is not merely “more images,” but more independent and deployment-relevant examples with trustworthy labels.

2. M4b and M5 are not interchangeable tests

The saved M4b camera set contained 30 frames with one cube of each color in view. Each predicted class ID appeared in all 30 frames, giving a 100% per-class frame-hit coverage result—not a ground-truth accuracy, recall, or AP result. The engine also produced more detections than frames: 91 blue, 52 green, and 34 red. The report explicitly notes color-confusable background and overlapping detections. [\[M4b evidence · M5 evidence\]](#)

Context	What it shows	Observed result
M2 Roboflow validation	training-domain validation	9 images, 21 instances, mAP@0.5 0.954
M4b saved camera frames	engine on an earlier robot-camera set	30 frames; blue/green/red frame-hit 100%/100%/100%
M5c2 live, sticker off, conf 0.50	current ROS node plus depth filter	0/460 kept detections on cubes-in-frame
M5c2 diagnostic, conf 0.25	lower threshold experiment	2/439 keeps, green only

WHY FRAME-HIT COVERAGE IS NOT RECALL

“A class appeared somewhere in every frame” does not match predictions to labelled objects. Extra boxes, wrong locations, duplicate boxes, and missed individual cubes can coexist with a 100% frame-hit rate. Recall requires ground-truth instances and a matching rule such as class agreement plus sufficient IoU.

3. What changed in the live room domain?

The M3d-revived analysis describes a downward JetRover camera at roughly 60–80 cm, with cube faces around 50–80 pixels, while the original Roboflow images are close-up views with much larger cube faces and cleaner backgrounds. The room also adds floor texture, lighting variation, distractors, and a different camera angle. These are plausible domain-gap mechanisms; they are not excuses until a controlled experiment supports them.

Scale Small faces occupy fewer pixels and can produce weaker features.	Viewpoint A downward room view is not the same geometry as a close-up training view.	Context Floor texture and colored non-cubes introduce new visual patterns.
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These factors can interact. A smaller cube has fewer edge and texture pixels; resizing to the fixed model canvas cannot recreate information the camera never captured. Different context can also change which candidate receives the strongest class score. This is why changing only the confidence threshold may expose symptoms without addressing the input mismatch.

4. Localize the failure in the pipeline

Do not jump directly from “no published box” to “geometry filter is broken.” The current node has separate stages: YOLO produces candidates, `_decode_yolov5_output()` applies confidence/NMS, then `compute_geometry()` and `decide()` inspect valid candidates. If the model never proposes a red or blue box, a later depth filter has nothing to evaluate.

Last evidence observed	First boundary to inspect
No synchronized callback/output messages	topics, timestamps, QoS, depth guard, node state
Model tensor exists but no row passes confidence	input preprocessing, artifact, domain, class scores, threshold
Post-confidence boxes exist but disappear in NMS	coordinates, class IDs, IoU, duplicate behavior
Post-NMS candidates exist but published list is empty	depth alignment, geometry statistics, decision policy

This is an application of a familiar debugging idea: instrument boundaries and locate the first stage where expected evidence disappears before changing code.

A published empty array is an **end-to-end symptom**, not a location. To blame the model, capture raw class scores before confidence filtering. To blame NMS, compare pre- and post-NMS boxes. To blame geometry, record post-NMS candidates and each geometry rejection reason. Without those intermediate observations, several explanations remain compatible with the same final output.

Interactive: choose the evidence-based diagnosis

If candidates exist but geometry rejects them as flat, inspect depth/box geometry and thresholds.

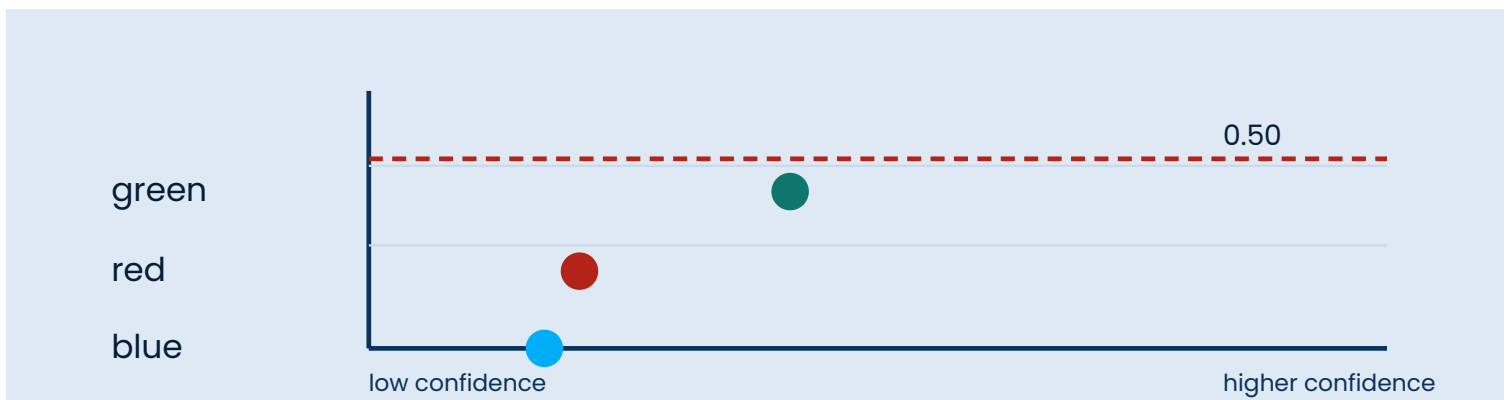
5. What the geometry evidence says

In the live empty bag, the node processed 439 messages and kept zero. The report records cumulative diagnostic totals of 991 `flat rejects` and one `aspect reject` across candidates seen during that run; these are not a one-to-one bag-message count. Zero kept detections is the observed empty-scene result. It is consistent with the filter rejecting candidates, but the current unvalidated depth-sign predicate means it does not by itself prove physically correct floor rejection. In the cubes bag at confidence 0.50, the recorded pipeline did not produce usable full-cube boxes for red or blue.

6. Why lowering confidence was a diagnostic, not a fix

The sticker-off re-run at `conf=0.25` produced only 2 keeps out of 439 messages, both `green_cube`. Red and blue stayed at zero. A lower threshold can reveal weak candidates, but it also expands the false-positive search space. This experiment narrowed the diagnosis: threshold alone is insufficient for the missing classes.

In metric language, lowering a threshold often increases recall but reduces precision. It is useful for asking “does any weak model evidence exist?”, but the production threshold must be chosen against labelled positives and negatives, not against one desired demo outcome.



The exact M5c2 experiment unlocked green at 0.25 but not red or blue. The diagram is explanatory; the counts above are the observed evidence.

7. The sticker hypothesis was tested and rejected

The M5c2 re-test removed the dimming sticker from the camera. The zero-kept result at confidence 0.50 remained: 0/414 in the earlier cubes bag versus 0/460 sticker-off. The recorded brightness delta was only +0.82/255, within the experiment's noise interpretation. This is a good example of changing one suspected cause and then accepting what the evidence says.

More precisely, this result rejects the simple claim “the sticker alone caused the zero-keep outcome under these test conditions.” It does not prove that illumination never matters, because the two bags are not every possible lighting condition and removing a sticker may change more than one image property.

8. Facts, hypotheses, and controlled tests

Statement	Evidence status	Next discriminating observation
Zero candidates are published at conf 0.50 in the cubes bags.	Measured end-to-end fact.	Already established by bag analysis.
The sticker alone caused the failure.	Contradicted under the recorded re-test.	No priority unless new lighting evidence appears.
Room-domain appearance weakens red and blue proposals.	Plausible hypothesis.	Log raw per-class scores before threshold/NMS on labelled room frames.
The geometry sign defect causes every miss.	Not established; impossible when no YOLO box reaches it.	Compare post-NMS candidates with geometry decisions.

A **controlled test** changes one factor while holding the rest as stable as practical. It also records the exact model hash, input frames, parameters, software version, and expected observable. Without that provenance, two runs that look comparable may exercise different systems.

9. Correctness and speed are separate gates

The live report measured a publish/frame-loop rate of 15.21–15.86 Hz, below the M5 report's 25 Hz frame-loop target, while the project definition separately states a final inference target of at least 5 fps. Do not use “the node is fast enough” to imply “the node detects correctly.” A pipeline can satisfy one dimension and fail another.

10. Honest conclusion

The evidence supports a narrow conclusion: current deployment artifacts execute, the empty-scene run publishes zero kept detections, and the live room pipeline does not reliably retain cube detections at the operating threshold. The candidate evidence makes a positive-detection/domain-gap hypothesis plausible, but it does not prove the model is the only cause or fully clear ROS wiring and geometry. Positive-domain fine-tuning is therefore a testable next intervention, not a verified fix.

11. Check your understanding

Why can high validation mAP coexist with live failure?

1. ☐ Validation measures deployment images using a different confidence threshold only.
2. ☐ The live images can differ in scale, viewpoint, lighting, and background.
3. ☐ TensorRT always changes the class mapping after exporting the checkpoint.

What can the depth filter not do?

1. ☐ Create a candidate that YOLO never proposed.
2. ☐ Reject a candidate whose geometry fails configured checks.
3. ☐ Return a reason alongside its keep-or-reject decision.

What did the sticker-removal experiment teach?

1. ☐ The sticker caused misses because removing it restored every class.
2. ☐ The sticker altered confidence because the threshold experiment fully recovered.
3. ☐ The sticker was unsupported because removal left the outcome unchanged.

Published detections are empty. What observation best distinguishes weak model scores from geometry rejection?

1. ☐ The final topic frequency by itself.
2. ☐ Candidates and class scores before confidence/NMS, followed by post-NMS geometry decisions.
3. ☐ The checkpoint file creation timestamp.

12. Read-only evidence exercise

Open `evaluation/m5_live/report.md` and write one sentence for each row: “What does this result rule in or rule out?” Use the empty bag, cubes bag at 0.50, and cubes diagnostic at 0.25. Do not call a threshold experiment a successful deployment.

13. Explain it back

Explain this without reading: “Why is domain gap a plausible hypothesis here, and what prevents us from calling it proven?”

Point to at least two measured results, the geometry sign defect, and one experiment that would reduce uncertainty across model, preprocessing, ROS, and geometry boundaries.

What this lesson intentionally postpones

This lesson intentionally postpones dataset construction and fine-tuning acceptance design to Lesson 0008, export mechanics to Lesson 0005, and the depth predicate derivation to Lesson 0007. Its job is to localize uncertainty before choosing an intervention.

Primary references and next lesson

- Ultralytics performance-metrics guide — why validation measurements must be interpreted in context.

- Dive into Deep Learning: underfitting, overfitting, and model selection — generalization and validation foundations.
- Project README and the linked canonical documentation.
- Curriculum plan — scope, teaching contract, and project truth.
- Lesson 0005 — ONNX and TensorRT: follow one trained function through its deployment representations.

CONTINUE WITH THE TEACHING AGENT

Ask the agent to clarify anything uncertain, challenge your explanation with a counterexample, or assess your explain-back. A learning record should be created only after you demonstrate the idea in your own words.

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Learning system · Recognition of Different Colored Cubes · Technical teaching artifact. Current engineering status: M5 PARTIAL; learner review remains separate from artifact validation.